25It036
Dhyana

PRACTICAL : 1

AIM 1 : Vending machine (VendingMachine.java, main in this file)

(a) Define an enum Coin with constants ONE, TWO, FIVE, TEN.

(b) In main, set a snack price of 15 and a running total of 0; create a Scanner.

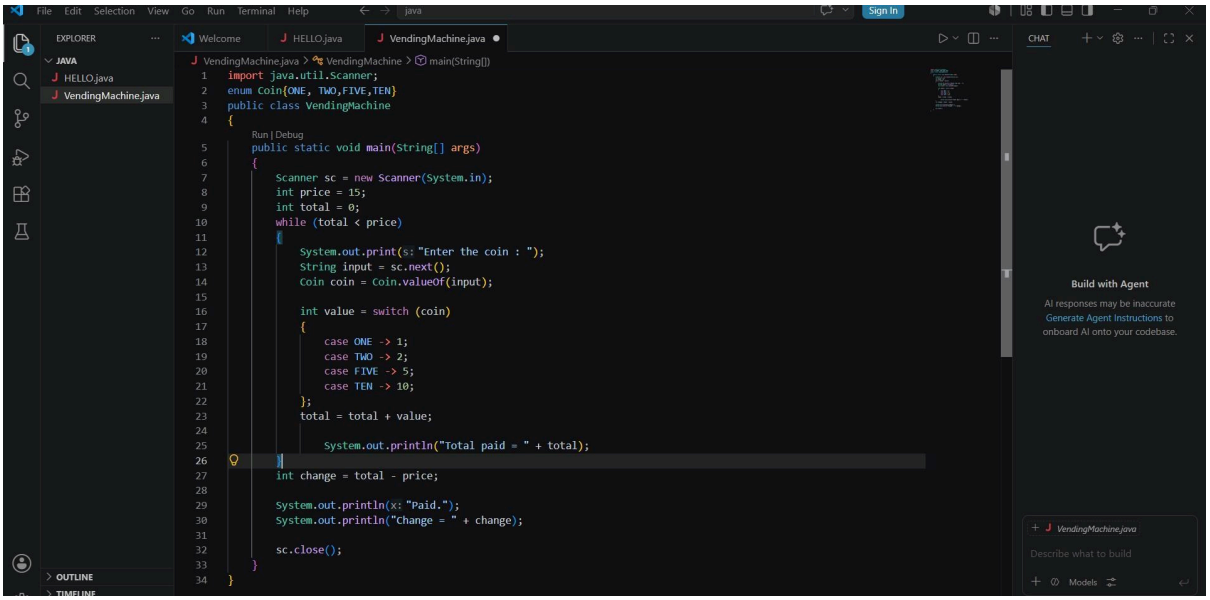
(c) Loop: read a coin name, use a switch expression to convert the Coin to its value (ONE→1, TWO→2, FIVE→5, TEN→10), add it to the total, and print the total so far.

(d) Stop the loop once the total reaches 15 or more.

(e) Print the change to return (total − 15).

Expected output: TEN then FIVE prints “Paid. Change: 0”; TEN then TEN prints “Paid. Change: 5”.

CODE :



```java
import java.util.Scanner;
enum Coin{ONE, TWO,FIVE,TEN}
public class VendingMachine
{
    Run | Debug
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        int price = 15;
        int total = 0;
        while (total < price)
        {
            System.out.print(s: "Enter the coin : ");
            String input = sc.next();
            Coin coin = Coin.valueOf(input);

            int value = switch (coin)
            {
                case ONE -> 1;
                case TWO -> 2;
                case FIVE -> 5;
                case TEN -> 10;
            };
            total = total + value;

                System.out.println("Total paid = " + total);
            }
        int change = total - price;

        System.out.println(x: "Paid.");
        System.out.println("Change = " + change);

        sc.close();
    }
}
```

OUTPUT :



AIM 2 : Toll booth (TollBooth.java, main in this file)

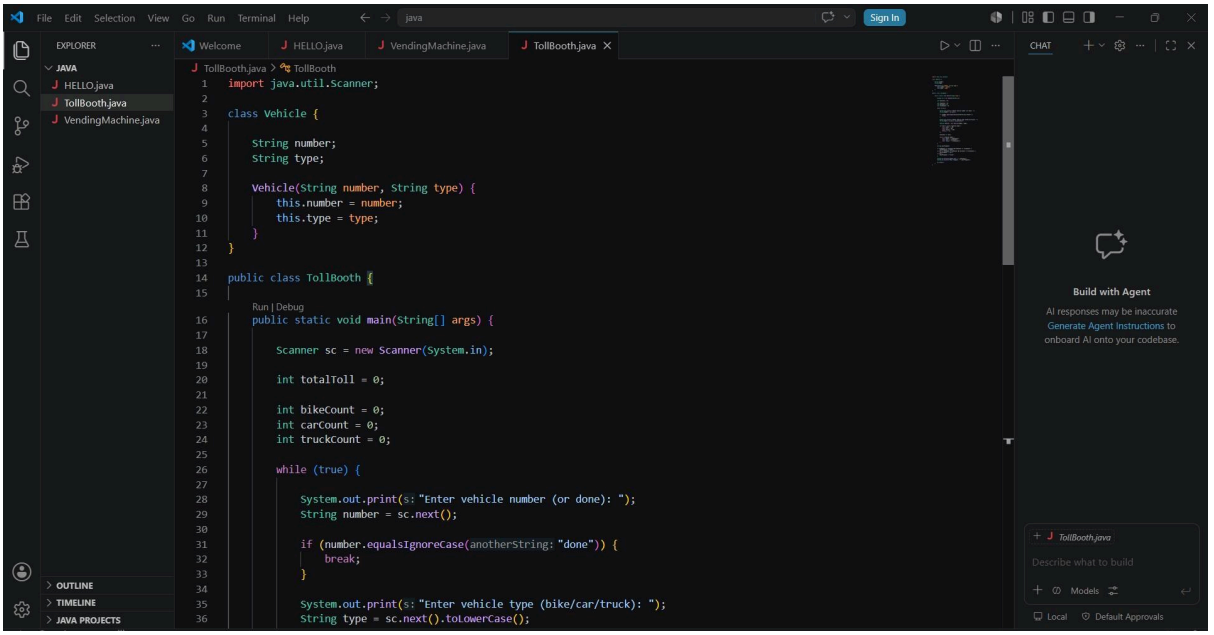
(a) Define a record Vehicle(String number, String type).

(b) In main, keep a running toll total and three counters (bike, car, truck). (c) Loop until the user types “done” for the number: build a Vehicle, use a switch expression on its type for the toll (bike→20, car→50, truck→150), add to the total, and increment that type’s counter.

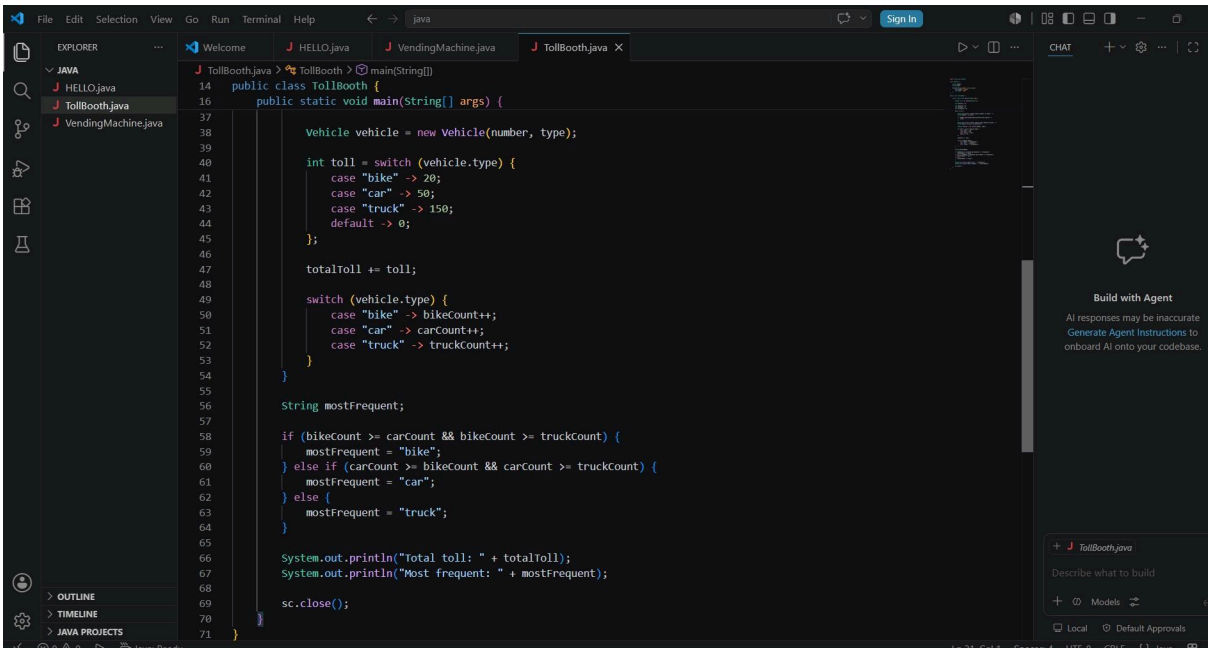
(d) After the loop, print the total toll and the type with the highest count.

Expected output: two cars + one bike → “Total toll: 120”, “Most frequent: car”.

CODE :



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```java
Vehicle vehicle = new Vehicle(number, type);

int toll = switch (vehicle.type) {
    case "bike" -> 20;
    case "car" -> 50;
    case "truck" -> 150;
    default -> 0;
};

totalToll += toll;

switch (vehicle.type) {
    case "bike" -> bikeCount++;
    case "car" -> carCount++;
    case "truck" -> truckCount++;
}
}

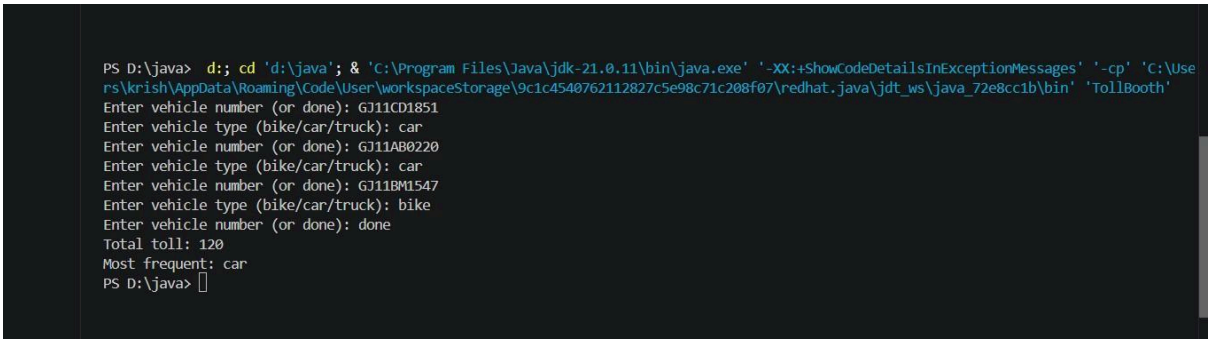
String mostFrequent;

if (bikeCount >= carCount && bikeCount >= truckCount) {
    mostFrequent = "bike";
} else if (carCount >= bikeCount && carCount >= truckCount) {
    mostFrequent = "car";
} else {
    mostFrequent = "truck";
}

System.out.println("Total toll: " + totalToll);
System.out.println("Most frequent: " + mostFrequent);

sc.close();
}
}
```

OUTPUT :



```
PS D:\java>  d:; cd 'd:\java'; & 'C:\Program Files\Java\jdk-21.0.11\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Use
rs\krish\AppData\Roaming\Code\User\workspaceStorage\9c1c4540762112827c5e98c71c208f07\redhat.java\jdt_ws\java_72e8cc1b\bin' 'TollBooth'
Enter vehicle number (or done): GJ11CD1851
Enter vehicle type (bike/car/truck): car
Enter vehicle number (or done): GJ11AB0220
Enter vehicle type (bike/car/truck): car
Enter vehicle number (or done): GJ11BM1547
Enter vehicle type (bike/car/truck): bike
Enter vehicle number (or done): done
Total toll: 120
Most frequent: car
PS D:\java>
```

AIM 3 :

2. Toll booth (TollBooth.java, main in this file)

(a) Define a record Vehicle(String number, String type).

(b) In main, keep a running toll total and three counters (bike, car, truck). (c) Loop until the user types “done” for the number: build a Vehicle, use a switch expression on its type for the toll (bike→20, car→50, truck→150), add to the total, and increment that type’s counter.

(d) After the loop, print the total toll and the type with the highest count.

Expected output: two cars + one bike → “Total toll: 120”, “Most frequent: car”.

CODE :

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```java
import java.util.Random;
import java.util.Scanner;

enum Move {ROCK, PAPER, SCISSORS, LIZARD, SPOCK}
public class RPSLS {
    static int winner(Move player, Move computer) {
        if (player == computer) {
            return 0;
        }
        return switch (player) {

            case ROCK ->
                (computer == Move.SCISSORS || computer == Move.LIZARD) ? 1 : -1;

            case PAPER ->
                (computer == Move.ROCK || computer == Move.SPOCK) ? 1 : -1;

            case SCISSORS ->
                (computer == Move.PAPER || computer == Move.LIZARD) ? 1 : -1;

            case LIZARD ->
                (computer == Move.SPOCK || computer == Move.PAPER) ? 1 : -1;

            case SPOCK ->
                (computer == Move.SCISSORS || computer == Move.ROCK) ? 1 : -1;
        };
    }

    Run | Debug
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        Random random = new Random();

        Move[] moves = Move.values();

        int playerScore = 0;
```

```java
    public class RPSLS {
        public static void main(String[] args) {
            int computerScore = 0;

            for (int round = 1; round <= 5; round++) {

                System.out.println("\nRound " + round);

                Move computer = moves[random.nextInt(bound: 5)];

                System.out.print(s: "Enter your move (ROCK, PAPER, SCISSORS, LIZARD, SPOCK): ");

                Move player = Move.valueOf(sc.next().toUpperCase());

                int result = winner(player, computer);

                System.out.println("Player   : " + player);
                System.out.println("Computer : " + computer);

                if (result == 1) {
                    System.out.println(x: "You win this round.");
                    playerScore++;
                }
                else if (result == -1) {
                    System.out.println(x: "Computer wins this round.");
                    computerScore++;
                }
                else {
                    System.out.println(x: "Round is a tie.");
                }
            }

            System.out.println(x: "\nFinal Score");
            System.out.println("You      : " + playerScore);
            System.out.println("Computer : " + computerScore);

            if (playerScore > computerScore) {
```

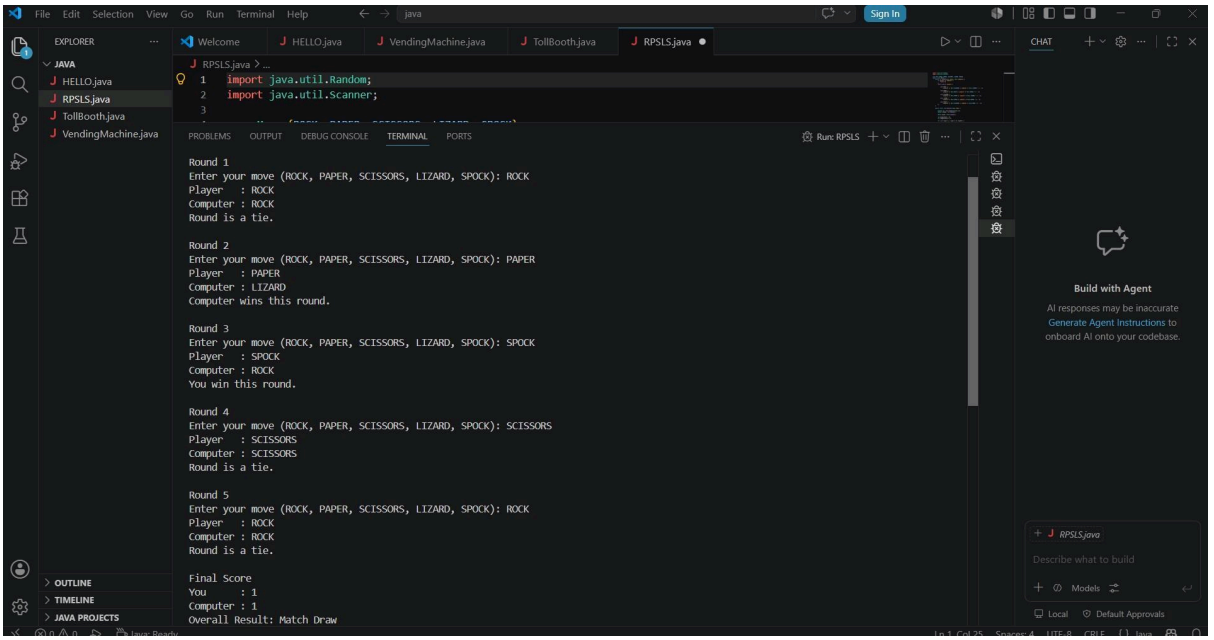
```java
                System.out.println(x: "Round is a tie.");
            }
        }

        System.out.println(x: "\nFinal Score");
        System.out.println("You      : " + playerScore);
        System.out.println("Computer : " + computerScore);

        if (playerScore > computerScore) {
            System.out.println("Overall Winner: You win " + playerScore + "-" + computerScore);
        }
        else if (computerScore > playerScore) {
            System.out.println("Overall Winner: Computer wins " + computerScore + "-" + playerScore);
        }
        else {
            System.out.println(x: "Overall Result: Match Draw");
        }

        sc.close();
    }
}
```

OUTPUT :



LEARNING OUTCOME :

In this practical, I learned how to set up the Java Development Kit (JDK) and configure Visual Studio Code for Java programming. I understood the basic structure of a Java program and learned how to write, save, compile, and run Java programs successfully. I also learned how to use the Scanner class to take user input and System.out.println() to display output. Additionally, I gained experience in identifying and correcting basic compilation and runtime errors, organizing Java source files, running programs through the terminal, and understanding the basics of version control using Git for managing Java projects.